HANDWRITTEN BUG EXPLANATIONS

Bug 1: SQL AND/OR Precedance issue

Location: TaskRepository.java — SearchTasks() query

How I discovered: while reviewing the SQL query, I noticed that AND or OR conditions where not grouped with parentheses.

Root Cause: → SQL gives higher precedence to AND than OR. Because of this, the archived and status conditions were not applied correctly to both title and description searches.

fix: I grouped the title and description conditions using parentheses.

Fixed logic: archived = FALSE, AND (title matches OR Description matches) AND (status matches).

Why this fix: This ensure that archived tasks are excluded and the status filter is correctly applied to both title and description.

Bug 2: Artificial API delay.

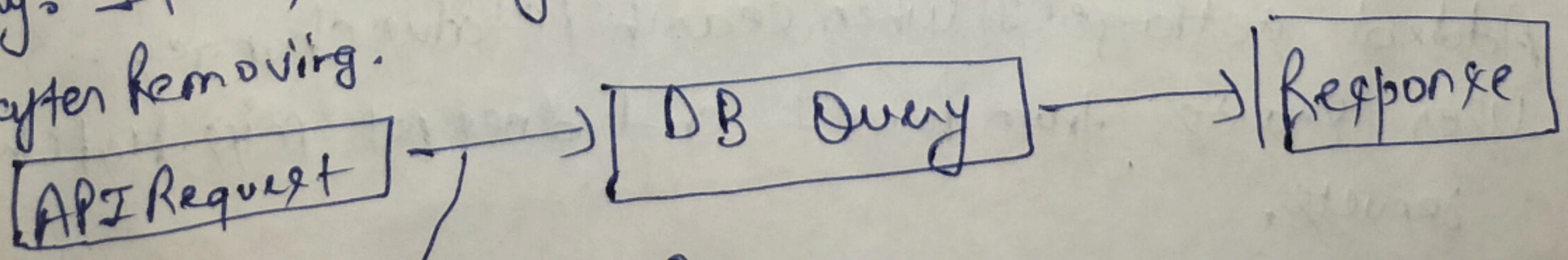
Location: TaskController.java

Root Cause: Thread.sleep() was adding unnecessary delay before executing DB query.

fix: Remove Artificial Thread.sleep() logic

Why: It was slowing API responses without any functional need.

→ after Removing.



but in the ~~Before~~ Thread.sleep() apply here.

Bug 3: Invalid Pagination Input
Location: TaskControllers.java
Root Cause: Page and page size were used without validating their values.
fix: Added validation [Page] >=1 pageSize ⟶ 1 to 100
why: Prevent invalid pagination and returns a clear bad Request.

Bug 4: Invalid Status Handling Location: TaskController.java
Root Cause: TaskStatus.valueOf() throws an exceptions for an invalid status.
fix: Added try-catch and returns 400 bad Request
Why: Valid user Input is handled gayely instead of causing an unexcepted server error.

Bug 5: Loading State Not Reset Location: fronterd/src/hooks/useTasks
Root Cause: setLoading(false) was missing from the catch block.
fix: Added setLoading(false) when the API request fails.
Why: The UI No longer remains stuck on "Loading tasks..." after an API error.

[API Request] ⟶ [Success?] —yes→ Loading false
[Success?] —NO→ set Loading(false ⟶ show Error.

Bug 6: Pagination Reset: Pagination Not Reset after filters
Location: fronterd/src/App.jsx
Root Cause: Page Number remained unchanged after search or status change.
fix: Added setPage(1) when search/status changes.
Why: User starts from page 1 and does not miss Matching results.

[Page 5] ⟶ [change Search] ⟶ setPage(1) ⟶ [Page 1 Results]
[Correct Results] ←